RIGHT EXACTNESS OF THE CHOW GROUP SEQUENCE ASSOCIATED WITH AN ENVELOPE, WITH THE CYCLE-LEVEL LIFTING CLAIM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Proper pushforward and flat pullback on cycles and cycle classes. Same chapter chow.tex, the sections labelled section-proper-pushforward and section-push-pull, both invoked at proof line 3610. Supplies that pushforward is defined for the proper morphisms f, p and q; the whole lifting argument stays in the counted span.
- Compatibility of proper pushforward with composition. Same chapter chow.tex, label lemma-compose-pushforward, invoked at proof lines 3614 and 3659. Gives the vanishing of the composite f_* p_* - f_* q_* and converts div(g_j) on W_j into div(f_j) on Z_j; the construction of beta' is in the counted span.
- Local finiteness of the images of a locally finite family. Same chapter chow.tex, label lemma-quasi-compact-locally-finite, invoked at proof line 3701. Makes the family of images f(W_j) locally finite on X so the assembled cycle gamma is well defined.

Source reference. Original lemma 02R5: sources/stacks-chow/pinned-original-source.tex, source lines 1924–1924. Navigation pointer for \ref{lemma-compose-pushforward} used inside the counted span (line 3614); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R2: sources/stacks-chow/pinned-original-source.tex, source lines 1847–1847. Navigation pointer for \ref{lemma-quasi-compact-locally-finite} used inside the counted span (line 3701); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02R3: sources/stacks-chow/pinned-original-source.tex, source lines 1880–1880. Navigation pointer for \ref{section-proper-pushforward} used inside the counted span (line 3610); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RF: sources/stacks-chow/pinned-original-source.tex, source lines 2393–2393. Navigation pointer for \ref{section-push-pull} used inside the counted span (line 3610); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3594); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $f : Y \to X$ be an envelope. Then we have an exact sequence*

$$\mathrm{CH}_k(Y \times_X Y) \xrightarrow{p_* - q_*} \mathrm{CH}_k(Y) \xrightarrow{f_*} \mathrm{CH}_k(X) \to 0$$

for all $k \in \mathbf{Z}$. Here $p, q : Y \times_X Y \to Y$ are the projections.

Proof. Since f is an envelope, f is proper and hence pushforward on cycles and cycle classes is defined, see Sections 02R3 and 02RF. Similarly, the morphisms p and q are proper as base changes of f. The composition of the arrows is zero as $f_* \circ p_* = (p \circ f)_* = (q \circ f)_* = f_* \circ q_*$, see Lemma 02R5.

Let us show that $f_* : Z_k(Y) \to Z_k(X)$ is surjective. Namely, suppose that we have $\alpha = \sum n_i[Z_i] \in Z_k(X)$ where $Z_i \subset X$ is a locally finite family of integral closed subschemes. Let $x_i \in Z_i$ be the generic point. Since f is an envelope and hence completely decomposed, there exists a point $y_i \in Y$ with $f(y_i) = x_i$ and with $\kappa(y_i)/\kappa(x_i)$ trivial. Let $W_i \subset Y$ be the integral closed subscheme with generic point y_i. Since f is closed, we see that $f(W_i) = Z_i$. It follows that the family of closed subschemes W_i is locally finite on Y. Since $\kappa(y_i)/\kappa(x_i)$ is trivial we see that $\dim_\delta(W_i) = \dim_\delta(Z_i) = k$. Hence $\beta = \sum n_i[W_i]$ is in $Z_k(Y)$. Finally, since $\kappa(y_i)/\kappa(x_i)$ is trivial, the degree of the dominant morphism $f|_{W_i} : W_i \to Z_i$ is 1 and we conclude that $f_*\beta = \alpha$.

Since $f_* : Z_k(Y) \to Z_k(X)$ is surjective, a fortiori the map $f_* : \mathrm{CH}_k(Y) \to \mathrm{CH}_k(X)$ is surjective.

Let $\beta \in Z_k(Y)$ be an element such that $f_*\beta$ is zero in $\mathrm{CH}_k(X)$. This means we can find a locally finite family of integral closed subschemes $Z_j \subset X$ with $\dim_\delta(Z_j) = k + 1$ and $f_j \in R(Z_j)^*$ such that

$$f_*\beta = \sum (Z_j \to X)_* \mathrm{div}(f_j)$$

as cycles where $i_j : Z_j \to X$ is the given closed immersion. Arguing exactly as above, we can find a locally finite family of integral closed subschemes $W_j \subset Y$ with $f(W_j) = Z_j$ and such that $W_j \to Z_j$ is birational, i.e., induces an isomorphism $R(Z_j) = R(W_j)$. Denote $g_j \in R(W_j)^*$ the element corresponding to f_j. Observe that $W_j \to Z_j$ is proper and that $(W_j \to Z_j)_*\mathrm{div}(g_j) = \mathrm{div}(f_j)$ as cycles on Z_j. It follows from this that if we replace β by the rationally equivalent cycle

$$\beta' = \beta - \sum (W_j \to Y)_* \mathrm{div}(g_j)$$

then we find that $f_*\beta' = 0$. (This uses Lemma 02R5.) Thus to finish the proof of the lemma it suffices to show the claim in the following paragraph.

Claim: if $\beta \in Z_k(Y)$ and $f_*\beta = 0$, then $\beta = \delta + p_*\gamma - q_*\gamma$ in $Z_k(Y)$ for some $\gamma \in Z_k(Y \times_X Y)$. Namely, write $\beta = \sum_{j \in J} n_j[W_j]$ with $\{W_j\}_{j \in J}$ a locally finite family of integral closed subschemes of Y with $\dim_\delta(W_j) = k$. Fix an integral

closed subscheme $Z \subset X$. Consider the subset $J_Z = \{j \in J : f(W_j) = Z\}$. This is a finite set. There are three cases:

(1) $J_Z = \emptyset$. In this case we set $\gamma_Z = 0$.
(2) $J_Z \neq \emptyset$ and $\dim_\delta(Z) = k$. The condition $f_*\beta = 0$ implies by looking at the coefficient of Z that $\sum_{j \in J_Z} n_j \deg(W_j/Z) = 0$. In this case we choose an integral closed subscheme $W \subset Y$ which maps birationally onto Z (see above). Looking at generic points, we see that $W_j \times_Z W$ has a unique irreducible component $W_j' \subset W_j \times_Z W \subset Y \times_X Y$ mapping birationally to W_j. Then $W_j' \to W$ is dominant and $\deg(W_j'/W) = \deg(W_j/W)$. Thus if we set $\gamma_Z = \sum_{j \in J_Z} n_j [W_j']$ then we see that $p_*\gamma_Z = \sum_{j \in J_Z} n_j [W_j]$ and $q_*\gamma_Z = \sum_{j \in J_Z} n_j \deg(W_j'/W)[W] = 0$.
(3) $J_Z \neq \emptyset$ and $\dim_\delta(Z) < k$. In this case we choose an integral closed subscheme $W \subset Y$ which maps birationally onto Z (see above). Looking at generic points, we see that $W_j \times_Z W$ has a unique irreducible component $W_j' \subset W_j \times_Z W \subset Y \times_X Y$ mapping birationally to W_j. Then $W_j' \to W$ is dominant and $k = \dim_\delta(W_j') > \dim_\delta(W) = \dim_\delta(Z)$. Thus if we set $\gamma_Z = \sum_{j \in J_Z} n_j [W_j']$ then we see that $p_*\gamma_Z = \sum_{j \in J_Z} n_j [W_j]$ and $q_*\gamma_Z = 0$.

Since the family of integral closed subschemes $\{f(W_j)\}$ is locally finite on X (Lemma 02R2) we see that the k-cycle

$$\gamma = \sum\nolimits_{Z \subset X \text{ integral closed}} \gamma_Z$$

on $Y \times_X Y$ is well defined. By our computations above it follows that $p_*\gamma_Z = \beta$ and $q_*\gamma_Z = 0$ which implies what we wanted to prove. □